

Geophysics 3A — Week 9 Workshop

Geothermics: Transient Heat Flow

May 27, 2026

Workshop Objectives

Students will:

- Understand how thermal diffusivity controls timescales.
- Compare thermal and chemical diffusion processes.
- Interpret half-space and plate cooling models.
- Use Fourier series to conceptualize thermal evolution.

0:00–0:20 Diffusivity and Thermal Timescales

- Review $\kappa = k/(\rho c_p)$.
- Emphasize modest variation in crustal rocks.
- Contrast sediments, rock, and ice.

0:20–0:45 Thermal Memory in Earth Materials

- Penetration depth scaling: $z \sim \sqrt{\kappa t}$.
- Borehole paleothermometry vs ice-core records.
- Depth–resolution tradeoffs.

0:45–1:15 Thermal vs Chemical Diffusion

- Both obey $\sqrt{Dt}$ scaling.
- Chemical diffusion often multi-component.
- Zoning persistence vs thermal smoothing.

1:15–1:25 Break

1:25–2:00 Half-Space Cooling

- Assumptions and boundary conditions.
- Growth of thermal boundary layer.
- Oceanic lithosphere as a thermal clock.

2:00–2:30 Plate Cooling and Fourier Series

- Fixed plate thickness as boundary condition.
- Fourier series as superposition of thermal modes.
- Mode decay times and geological interpretation.

2:30–3:00 Stefan Problems in Geology

- Latent heat as effective heat capacity.
- Cooling times of dikes vs sills.
- When analytic solutions break down.